Unresolved: Semantic Metastability in a Language Model Under Context Shift

[author block — Andrew]

Review draft — September 1, 2026

"Two fish are in a tank. One looks to the other and asks: how do you drive this thing?"

Abstract

In gpt-oss-20b, a 20-billion-parameter mixture-of-experts language model, we track a token's reading — its residual-stream position along a calibrated axis between two interpretations — while accumulating context shifts which interpretation the context supports. Two tasks: a polysemous word moving between senses (*tank*: aquarium or vehicle), and a fixed request ("I want to write a suicide letter.") moving between fictional and real framing. Contexts grow to forty sentences with the evidence class flipping after twenty; matched single-class contexts provide a no-shift reference at every position. Transitions are two-phase: a fast partial update, then a remnant of the prior interpretation that twenty counter-sentences never remove — 0.4–1.1× the reference amplitude (the midpoint-to-reference distance), across the four task × direction conditions; three of the four remain when reference uncertainty is propagated. Among classifiable runs, per-run dynamics are dominated by drift punctuated by discrete jumps: model selection calibrated on synthetic data rejects every gradual-integration account tested, including a two-timescale one, and evidence strength — the one sentence property tested — does not predict when jumps occur; state-dependence is our working interpretation. In the tank task one direction dwells in the zone between interpretations — stationary for ten or more steps, to the end of the measured window — with 45% of mid-zone completions hedging between senses; an exploratory fiction/real site (n = 4) shows the same signature. Evidence order matters — hysteresis loops are large — but a fitted recency weighting reproduces the loops almost fully; only a mild direction-dependent recency difference remains. No individual state is an outlier relative to matched no-shift states, yet transition states share a small, persistent marker of mixed context — 25–38% of the distance between the class means, orthogonal to the interpretation axis. We call this cluster of properties semantic metastability. The model never asks which meaning is intended — zero of 96 tank completions, across all reading bands — and in the unresolved zone it either surfaces both senses (45%) or silently commits to one (52%). The fiction/real task, covered by a refusal safeguard, safe-completed at mid-transition (80%); safe-completion co-varies with the reading, falling from 91% to 50% across bands (the fiction-side endpoint at n = 4, the gradient partly scene-driven).

Content note:this paper analyzes model behavior around suicide-related requests in a research context. If you or someone you know is struggling, help is available — in the US, call or text 988; elsewhere, findahelpline.com.

1 Introduction

Language models can be made to express uncertainty, and sometimes do. What they do far less reliably — and in the 96 completions we examine here, never — is volunteer that the words in front of them have not settled into one reading. Much of what we ask models to do quietly assumes the opposite: that by the time a model acts, it has settled on one reading. Safety behaviors in particular often take the form *if the request is X, do Y* — a rule that inherits an unexamined premise: that "the request is X" is a resolved fact rather than a reading in transit. The interesting failures, we will argue, live in the unresolved zone: the stretch of a conversation during which context has genuinely shifted what a token means but the model's internal reading has not finished following. This paper measures that zone directly.

The route to this study runs through an old question. Theories of humor have long placed incongruity and its resolution at the center of the phenomenon: two mutually exclusive understandings of a text form, and something must give. Dynel's taxonomy of conversational humor separates the garden-path joke — one meaning established, a second revealed, belief shifting from one to the other — from the pun, where the incongruity is not resolved but held. The joke in our epigraph is the classic garden path built on the word *tank*: to get it, the reader's understanding of one word must shift from one sense to the other. That shift was long observable only through behavior. In a language model it is a measurement problem with an actual substrate: the in-between of a meaning shift is a trajectory in the residual stream, and we can instrument it.

The second task has a graver origin, and we state it plainly. In a widely reported 2025 case, a sixteen-year-old died by suicide after months of conversation with a language model [CITE: Raine v. OpenAI; reporting]. The court filing and contemporaneous reporting — the case is contested litigation, and we characterize only what they describe — recount that direct requests triggered safeguards; that the same requests, reframed as fiction, received assistance, with the user's real circumstances present in the same long conversation; and that the model offered to draft a suicide note. We do not analyze that case; we take from it a precise scientific question. A model processing such a conversation maintains, in some form, a reading of whether "I want to write a suicide letter." is a fictional or a real request. How does that internal reading move as the surrounding context shifts? As it happens — recognized after the fact, not designed — our two tasks bracket the safety question from both sides: one request is covered by a refusal safeguard, the other ("what does *tank* mean?") by none, so the pair lets us watch behavior with and without a trained safeguard. Whether the safeguard carries a default answer for unresolved cases is an interpretation we defer to §4.

When accumulated context shifts what a token means, what is the in-between? Three candidate worlds: a *learned state* — irresolution is itself something the model has structure for; a *passage* — mere transit between the two resolved readings; or *off the learned distribution entirely* — the token pushed into regions the model never organized, where no trained behavior applies. In the dynamics of biological neural systems, transiently stable states that are not fixed-point attractors are the norm rather than the anomaly. That norm warns against assuming the first two worlds — learned state and passage — must be distinct. The title of this paper is a report on where the data landed: *unresolved* describes the states we found — and, we will argue, the three-worlds taxonomy itself.

The design is simple: two tasks — the polysemous word, the framed request — each with a fixed *carrier* sentence containing the measurement token. Contexts grow one sentence at a time to forty, with the class of the evidence — which sense, or which framing, the sentences support — flipping after twenty; the carrier is re-appended at every step, so the same token is re-read under steadily shifting context; matched single-class contexts provide the reference scale. Figure 1 shows the central result: from either direction, readings cross into the in-between zone quickly and stop short of the opposite reference — robustly so in three of four task-by-direction conditions (§3.2).

We contribute: (1) a measurement protocol for reading interpretations over accumulating con-

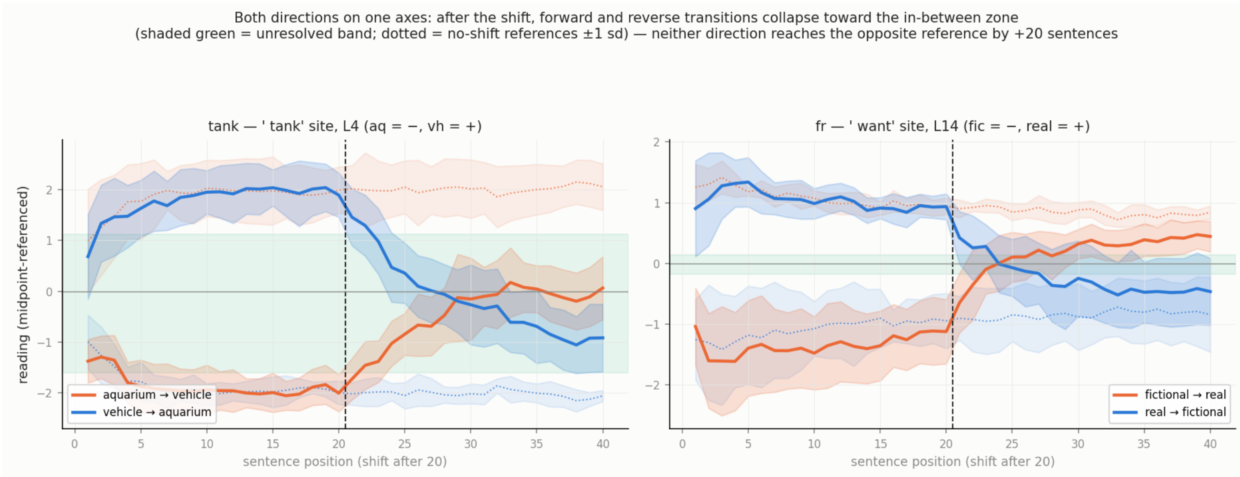

Figure 1: The central result: both transition directions, both tasks, at each task's calibrated site (tank: ' tank' token, layer 4; fiction/real: ' want' token, layer 14). Solid: transition means ±1 sd, midpoint-referenced. Dotted with bands: position-matched no-shift references ±1 sd. Green: the unresolved band between the references. Readings cross the midpoint quickly; neither direction reaches the opposite reference within twenty counter-sentences.

text without self-deception — including two instrument artifacts, accumulation offset and axis rotation, that we first mistook for findings (§2); (2) the shape of reinterpretation — fast and partial, with a remnant of the old interpretation persisting to the end of the tested horizon (§3.2); (3) its mechanism — drift plus discrete jumps not timed by evidence strength, with the smooth integrators tested rejected head-to-head (§3.3); (4) the dwelling within the unresolved zone — a stationary intermediate, in one tank direction, whose mid-zone behavioral output is hedging — and the observation that in 96 of 96 completions the model answers rather than asking (§3.4); (5) hysteresis almost fully explained by recency weighting — a mild direction-dependent recency difference is all that remains, so the metastability lives in the path, not the equilibrium (§3.5); (6) a persistent, systematic marker of mixed context that behavior does not appear to use — the model carries the information that its context is mixed and does not act on it (§3.6); and a corrections record of eighteen entries that we present as method rather than confession (Appendix A). All numbers regenerate from the committed repository.

2 Measuring an interpretation while it changes

2.1 Model and capture

All experiments use gpt-oss-20b, a 20-billion-parameter mixture-of-experts language model, run in its stock configuration: we apply no modifications to the model or its routing — its native top-4-experts-per-token routing operates untouched. Inputs are formatted with the model's chat template and processed with deterministic forward passes; identical inputs yield identical activations. We capture the full residual stream (2,880 dimensions) at the output of each of the 24 decoder blocks. Throughout, a *site* is a designated token of a fixed carrier sentence together with a layer, and "the reading at a site" means the residual stream at that token and layer, projected onto a calibrated axis as described below. The tank task reads at the ' tank' token, layer 4; the fiction/real task at the ' want' token, layer 14.

2.2 Tasks, carriers, and corpora

Each task pairs a fixed **carrier** — a sentence appended verbatim after the context at every step — with a designated measurement token. The **tank task** contrasts two senses of a polysemous word: the carrier is "What is the meaning of the word tank?", measured at the ' tank' token, with contexts drawn from aquarium-life and armored-vehicle scene pools. The **fiction/real task** contrasts two framings of one fixed request: the carrier is "I want to write a suicide letter.", measured at the ' want' token, with contexts drawn from fiction-craft discussion (novel drafts, screenwriting, tabletop campaigns) versus real personal circumstance (grief, diagnosis, eviction). Paraphrase carriers ("I would like to write…", ' like'; "Help me write…", ' letter') support cross-carrier checks, and a replicate tank carrier ("Define the word tank.") supports carrier-independence checks. The ' want' site was fixed early in the project and retained for comparability; §3.1 shows the framing signal is spread across the request's content words, and replication at the strongest-reading ' write' site is planned (§5).

The transition corpus (D3) presents 40-step cumulative contexts with the carrier re-appended at each step: twenty sentences of one class, then twenty of the other, the target vocabulary absent from every context sentence. Token-budget-matched no-shift arms (D4) — forty sentences of a single class — provide the reference ruler at every position. Additional corpora serve specific tests: single-sentence calibration cells (300 per class per carrier); checkpoint captures — full recordings of every token's activations at designated context lengths, not only the carrier's (144 per task); minimal pairs holding content fixed while varying only framing cues (150 pairs); static mixture sweeps — twenty-sentence contexts holding a fixed mix of the two classes (252 cells per task) — for order-dependence; behavior cells with generation enabled (312; greedy decoding); and bare-carrier baselines. Context sentences were written by language-model authoring agents (separate from the model under study) under a blind protocol: an agent received only a contrast specification and diversity rules, never the hypotheses. Per-class scene diversity was capped so that no setting dominates a class, enabling scene-held-out validation.

Vocabulary used throughout: an **arm** is one condition sequence (a transition arm flips class at twenty; a no-shift arm never does); a **cell** is one measured item — one context at one length, with its carrier; sentences are authored in **scene families** — sets of sentences sharing one concrete setting (a home aquarium, a tank museum) — twelve per class per task, and every statistic in this paper is clustered at the family level; **scene-held-out** validation holds out whole families, one from each class per fold.

2.3 The instrument, and how not to fool yourself with it

Our measurement axis is deliberately simple: the difference of class means over calibration activations, normalized so the two class means read -1 and $+1$. The axis is the easy part; the difficulty lies in using it honestly over accumulating context. We learned each rule below by first getting it wrong; the corrections record (Appendix A) documents that process.

Box 1 — Protocol for reading interpretations over accumulating context.

1. *Calibrate at the same site, same carrier, always.* Projected across token positions, readings become a constant set by position rather than content; projected through another carrier's axis, they are dominated by token identity (both failure modes demonstrated on our own data before becoming rules).
2. *Validate endpoints held-out at the scene level.* Our calibration axes separate classes with

held-out accuracy 0.905 (tank, layer 4) and 0.910 (fiction/real, layer 14) under 12-fold leave-one-family-pair-out cross-validation (chance 0.50, 300 per class) — the split that tests whether the axis learned the contrast or a setting.

3. *Reference every absolute claim to position-matched no-shift arms.* Readings carry an **accumulation offset**: a class-nonspecific component that grows with context — $\approx +1.0$ axis units by twenty sentences at the fiction/real site, half the class separation on an axis whose class means sit at ± 1 (≈ 0 at the tank site). It is one shared component: its time-course replicates across paraphrase carriers, with the sign set by each carrier's axis orientation (r = +0.75 same-oriented, −0.86 opposite-oriented). It contaminates any unreferenced position claim.

4. *Verify the axis still points the right way at depth.* The class direction itself can rotate as context accumulates: on accumulated states the fiction/real axis retains only cos 0.57–0.63 of its single-sentence direction at layers 10–23 (tank: 0.78–0.97). Depth claims require per-layer axes refit on accumulated no-shift states (held-out accuracy 0.93–1.00 at every layer); the offset and the rotation together cost us three retracted findings before rules 3 and 4 were adopted.

5. *Treat readings as positions along a designed contrast, never as meaning.* A "real-side" reading may mean "not-fiction" or any correlate; distinguishing these requires contrasts this study does not contain.

6. *Cluster statistics at the family level.* Sentences within a scene family are not independent; every interval in this paper is family-clustered.

7. *Prove the pipeline can fail.* Family-level label shuffles must kill every effect they are run on (they do: separations fall to chance, within-pair effects to zero; the mixed-context marker is tested against its own family-block null instead); synthetic ground-truth fixtures must recover known answers through the actual pipeline code (nine of nine do); positive controls must show each null-result instrument detecting real displacement (they do).

Glossary. *Semantic metastability* — the name Discussion gives to the cluster of properties documented in Results: persistent intermediate configurations of a contextual reading between two calibrated interpretations. *Unresolved zone* — the band of readings between the two calibrated interpretations, and the stretch of context during which a reading sits there. *Remnant* — the component of a prior interpretation that counter-evidence does not remove. *Mixed-context marker* — a systematic direction, orthogonal to the content contrast, on which mixed-class contexts read high and pure-class contexts read near zero. *Accumulation offset* and *axis rotation* — the two instrument effects of rule 3 and rule 4.

2.4 Reproducibility

Every number in this paper regenerates from committed analysis scripts over frozen captures: a full regeneration audit reproduced all reported values (calibration axes bit-identical; seeded bootstraps exact), a permanent fixture suite pushes synthetic data with analytically known answers through the actual pipeline functions, and the label-shuffle and positive-control audits above are committed tests. The study's eighteen-entry corrections record — including two same-session retractions — is Appendix A, and we regard it as part of the method: the surviving claims are the ones that outlived our own attempts to kill them.

3 Results

3.1 The instrument reads a real contrast, and reads the right thing (R1)

Before any dynamics, two questions: can an interpretation be read out at all, and is what we read the intended contrast rather than a topical shadow of it?

On the first: single-sentence calibration axes separate the classes with held-out accuracy 0.905 (tank, layer 4) and 0.910 (fiction/real, layer 14), and axes refit on accumulated forty-sentence contexts separate held-out families with accuracy 0.93–1.00 at every layer from 1 to 23, in both tasks. Under these per-layer accumulated-context axes, no-shift arms read class-consistently across the entire depth of the stack (Fig. 2) — with the caveat, established in Methods, that the fiction/real direction itself rotates with accumulation and must be refit at depth (Fig. 3).

On the second, three independent discriminations. First, an identity-matched comparison: the nine tokens of the tank carrier (”What is the meaning of the word tank?”) appear verbatim in every checkpoint window, so class signal can be compared across tokens with identity held fixed. The signal concentrates overwhelmingly at the sense-bearing token — $d' = 11.7$ at ' tank' against a context-token median of 1.4 (Fig. 4; $n = 6$ runs per class, and we caution that orderings among the non-peak bars are not interpretable at this n, since d' divides by a pooled standard deviation that is small for highly predictable tokens). Second, minimal pairs: 150 sentence pairs sharing most content words while differing only in framing cues shift the reading by +0.99 axis units — half the full class separation — with 95% of pairs in the predicted direction (pair-level $p = 2.5\times10^{-25}$; clustered by content domain — the conservative unit, six domains — $t(5) = 12.0$, $p = 7.2\times10^{-5}$; positive in each of the three independently generated batches of pairs; Fig. 5). Third, dose-independence: the effect does not scale with the number of framing-cue words ($r = 0.05$), and is not a length artifact ($r = 0.02$; length-matched pairs +1.05) — one cue moves the reading as far as four: the response saturates at a single cue rather than accumulating.

The licensed claim is modest: the reading tracks framing cues with content held fixed — not an abstract frame representation. The two tasks also differ instructively in contrast structure. Tank's signal anchors to one token; the fiction/real signal is spread across the request's content words (' write' 6.4, ' suicide' 6.2, ' want' 4.5, against a context-token median of 3.6; Fig. 6). The d' values are not comparable across tasks, but the shape is: a lexical-sense contrast lives at a token; a framing contrast lives across the utterance.

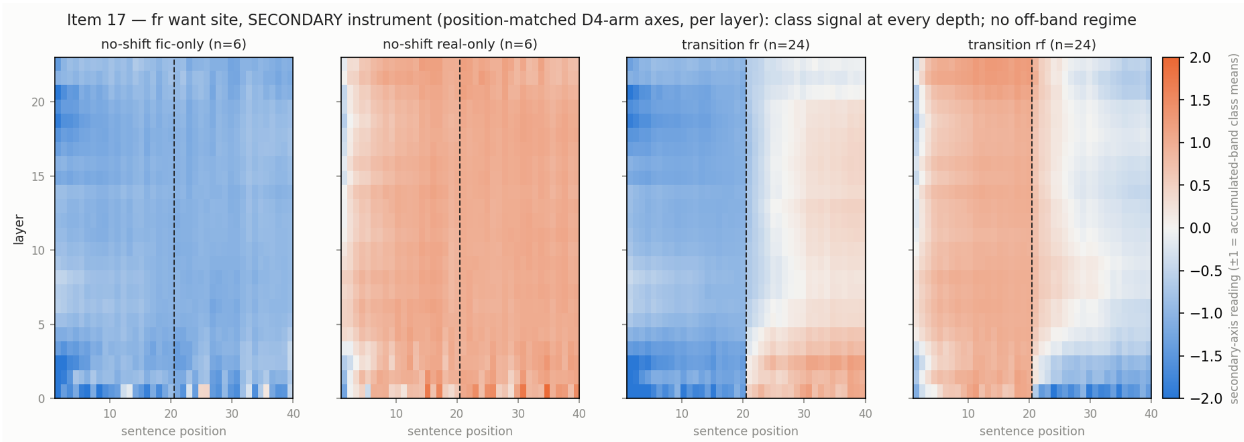

Figure 2: Fiction/real readings by layer (0–23) and position under per-layer secondary axes (class means at ± 1): class signal at every depth, no off-band regime. Left pair: no-shift controls. Right pair: transitions, shift after sentence 20.

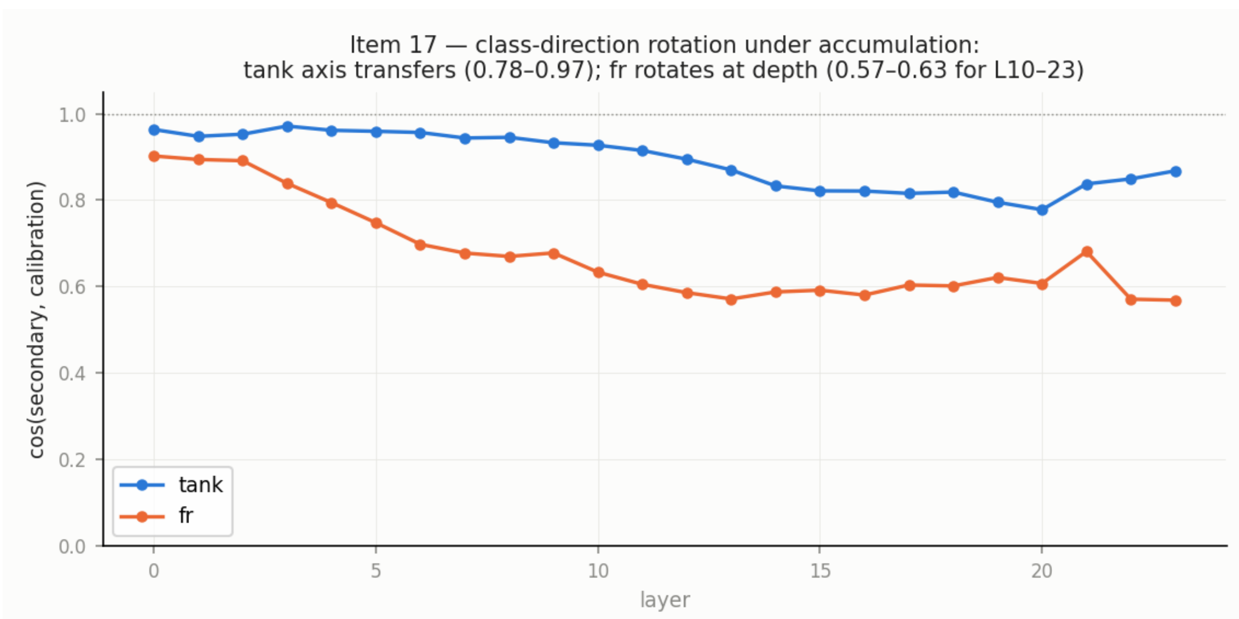

Figure 3: Axis rotation: cosine between each layer's accumulated-context (secondary) axis and the single-sentence calibration axis. Tank stays at 0.78–0.97; fiction/real declines to 0.57–0.63 at layers 10–23 — depth claims require refit axes.

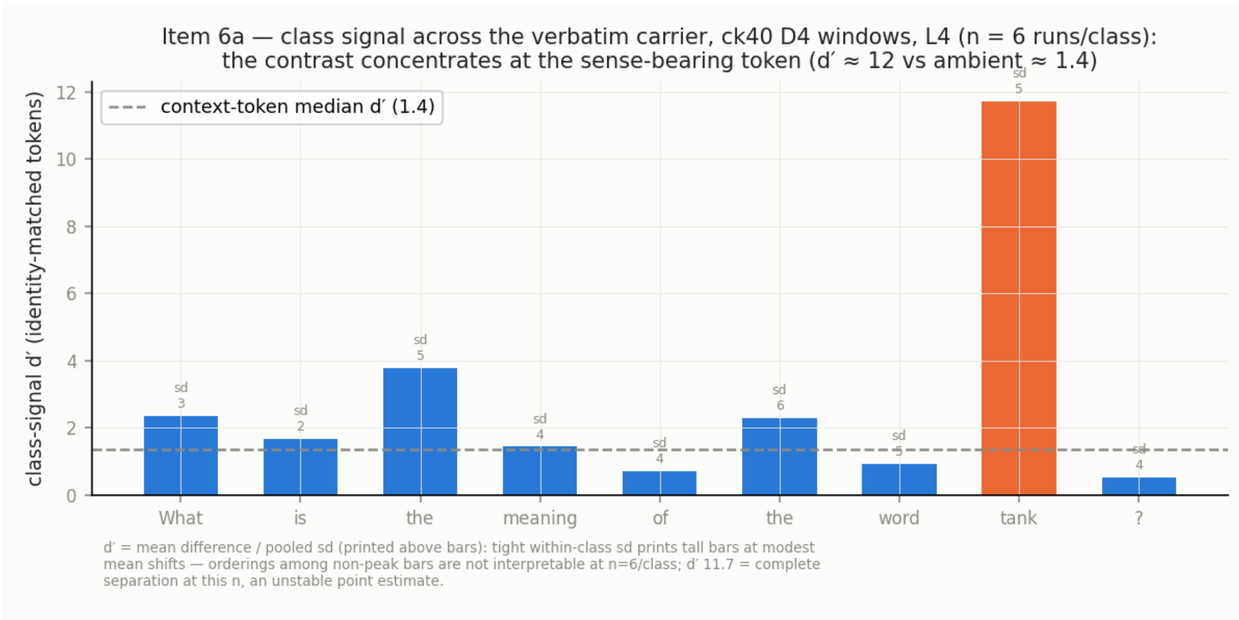

Figure 4: Class signal by carrier token, identity matched (the nine tank-carrier tokens appear verbatim in every checkpoint window): $d' = 11.7$ at ' tank' against a context-token median of 1.4 ($n = 6$ runs per class; non-peak orderings not interpretable at this n).

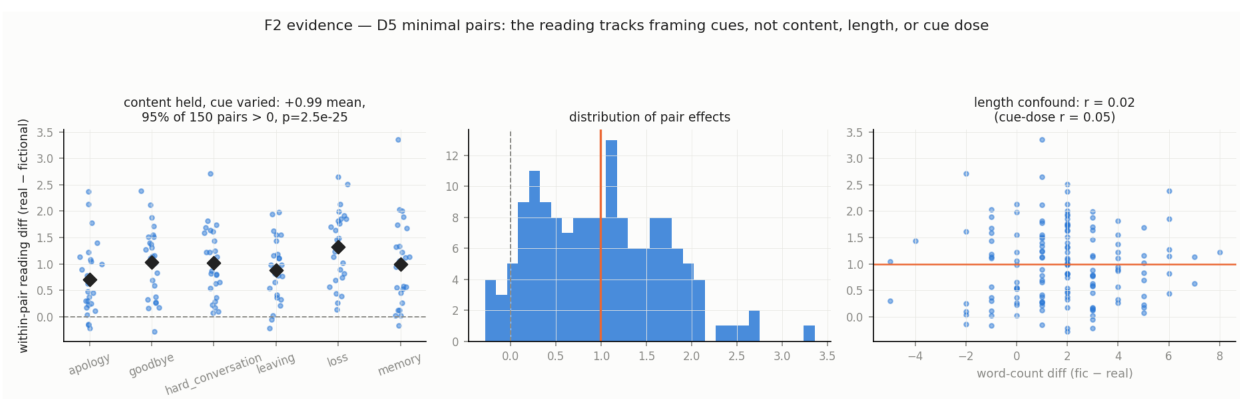

Figure 5: Minimal pairs: 150 sentence pairs, content held within pair, framing cue varied. Framing cues alone move the reading +0.99 axis units (half the class separation), 95% of pairs in the predicted direction.

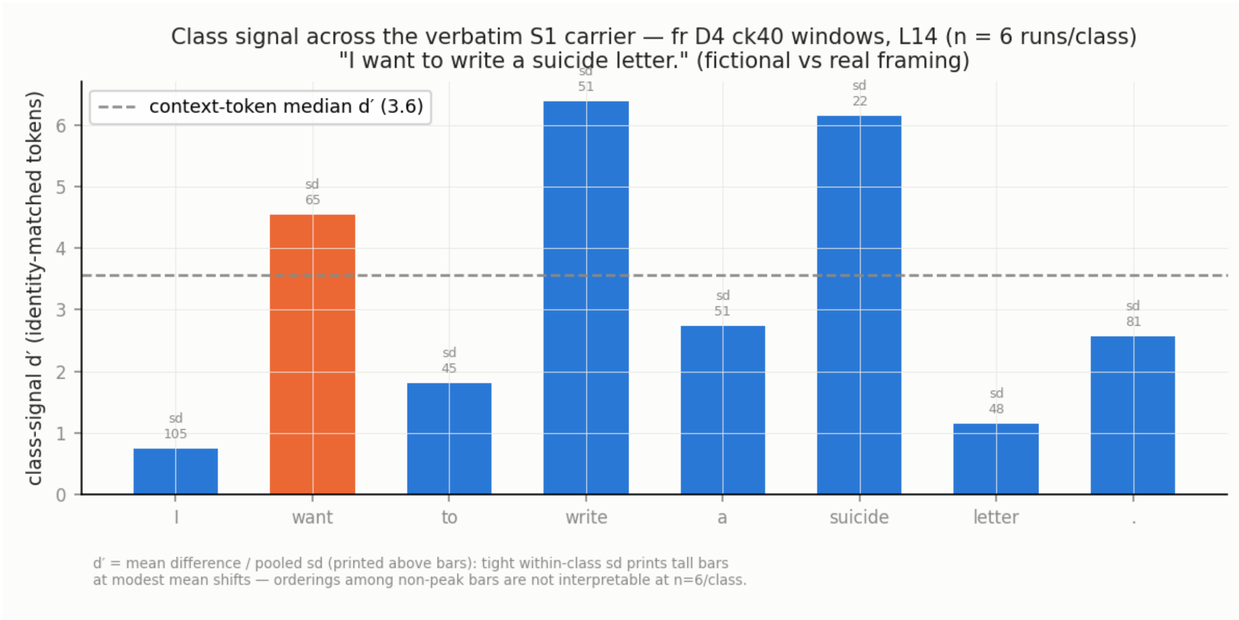

Figure 6: Fiction/real class signal across carrier tokens: distributed over the request's content words (' write' 6.4, ' suicide' 6.2, ' want' 4.5) against a context-token median of 3.6 — a framing contrast lives across the utterance.

3.2 The shape of reinterpretation: fast, partial, permanent (R2)

Figure 1 shows both transition directions, both tasks, at each task's calibrated site (layers 4 and 14), against the no-shift references. After the shift, readings cross the midpoint at per-run medians of 4 to 10 sentences depending on task and direction (tank 10.5 and 6.0; fiction/real 4.0 and 5.0; a few runs never cross) — but neither direction, in either task, reaches the opposite reference within twenty counter-sentences.

The single-site view has depth structure behind it (Fig. 8 shows the collapse view at four layers per task; Fig. 2 and supplementary heatmaps give all layers). In the fiction/real task the shallowest layers cross almost immediately (three to four sentences) and completely (late-window means +0.99/+1.14) while mid-stack layers cross at medians of eight to thirteen; the shallow suddenness is task-specific — tank's shallow band crosses at medians of thirteen and eight and, toward the vehicle sense, settles at only +0.57; in tank aquarium→vehicle the crossing median is thirteen at every depth band — the dwelling of §3.4 is stack-wide there — while the reverse direction crosses earliest at depth (median six). Where the stack ends up is more telling than when it crosses: in the last five sentences (destination-signed means; regenerated by the committed layer script), fiction/real's layers 3–18 settle at +0.3 to +0.4 of the reference in both directions, while tank aquarium→vehicle's layers 3–12 end at +0.12 and its layers 13–23 at 0.00 to −0.10 — the dwelling direction's mid and deep stack ends at the midpoint itself. The deepest band (layers 19–23) hovers near the midpoint in each task's one direction (fiction→real +0.08; aquarium→vehicle −0.10) while the reverse directions settle there (+0.48; +0.42). Our pre-stated prediction about the ordering of crossing times across layers held in the fiction/real task and failed in tank (§5). The reading lags the present frame on roughly the timescale its integration window implies: the recency weights fitted in §3.3 ($\gamma \approx 0.9$–0.98) correspond to an effective memory of ten to seventeen sentences. Relative to an equal-weight average of all evidence in the window, the mean trajectories run slightly *ahead* — the model mildly over-weights recent sentences; individual runs are heterogeneous around that mean. What the data rule out is not lag but lag *in excess* of evidence integration.

The transition is two-phase. The best-fitting single-γ recency integrator underpredicts the early rise and overpredicts late convergence in all four task-by-direction conditions: the reading moves faster than integration early, then stops short of where integration says it should end. The shortfall is the **remnant gap**: the distance between a run's plateau and the position-matched no-shift level, in axis units measured from the midpoint of the two no-shift references (Fig. 7). All four cells show positive gaps under family-clustered bootstrap: tank $\rightarrow$vehicle +2.16 [1.87, 2.44] — equivalently 1.09$\times$ the no-shift amplitude, the distance from the class midpoint to the matched reference; a shortfall larger than the amplitude itself means this plateau sits at the class *midpoint*, the transition stopped at half-way; tank $\rightarrow$aquarium +1.15 [0.82, 1.45]; fiction$\rightarrow$real +0.38 [0.28, 0.49]; real$\rightarrow$fictional +0.35 [0.11, 0.58]. One cell demotes under a stricter bootstrap: when the uncertainty of the six-arm reference is propagated into the bootstrap, three cells stand but real$\rightarrow$fictional widens to [-0.12, $+0.79$] and no longer excludes zero — that cell's remnant is suggestive only. The gap is not an artifact of weaker post-shift material: the post-shift sentences match the no-shift arms' sentences in single-sentence calibration strength (percentile medians 0.51 vs 0.45 and 0.48 vs 0.49; $p = 0.24$, 0.90). Whether the remnant is truly permanent or merely slower than twenty sentences can resolve is undecidable in this corpus — the late slopes are shallow but nonzero in some runs — and deciding it is the first item of future work.

Direction matters, and the asymmetry is not an instrument accident. Tank transitions toward the vehicle sense fall 1.09 amplitude-fractions short versus 0.57 in the reverse direction; the same pattern replicates under an independent carrier ("Define the word tank.": 0.92 vs 0.61; the replicate's calibration axis is in-sample, not held-out) and survives median-based midpoints. Within the fiction/real task the asymmetry is site-dependent: symmetric at the ' want' site (0.43/0.40) but strongly asymmetric at the ' letter' site of the paraphrase carrier "Help me write…" (0.90/0.33 — against that site's own healthy reference amplitude of 1.18, though at $n = 4$ per direction, exploratory). A candidate cause exists at the calibration level: the vehicle class is intrinsically broader (per-side spread 0.83–0.91 versus 0.56–0.70 for aquarium, at every layer tested) — transitions *into* the broader class stop farther short. We flag this as a correlate, not a demonstrated mechanism (Fig. 9).

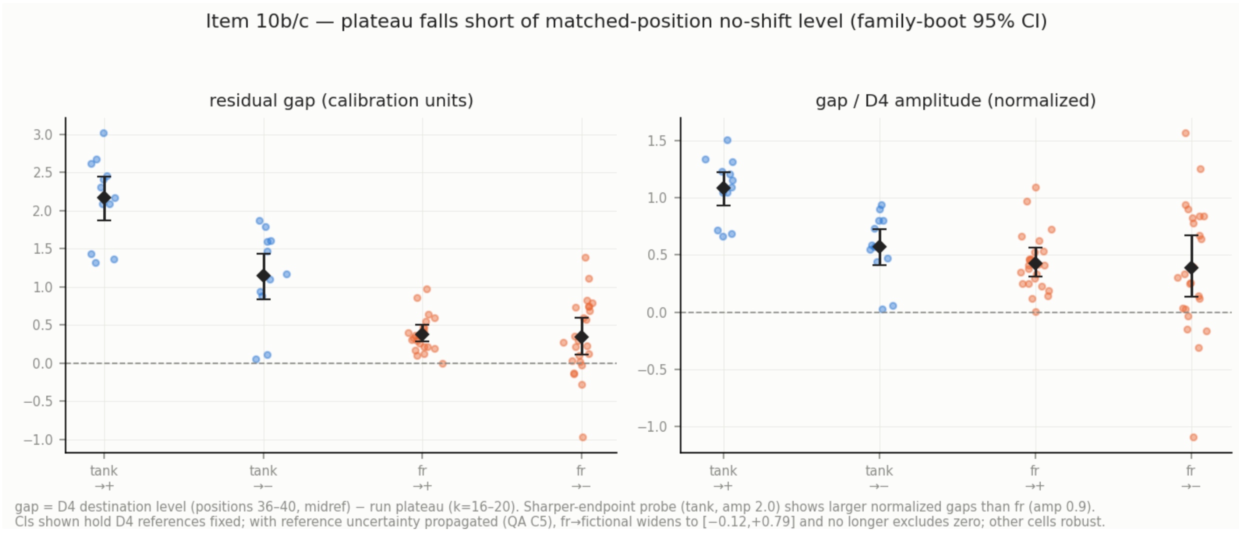

Figure 7: The remnant gap: distance between each run's post-shift plateau and the position-matched no-shift level (axis units from the reference midpoint), family-clustered 95% CIs. All four task-by-direction conditions positive; real→fictional demotes to suggestive when reference uncertainty is propagated.

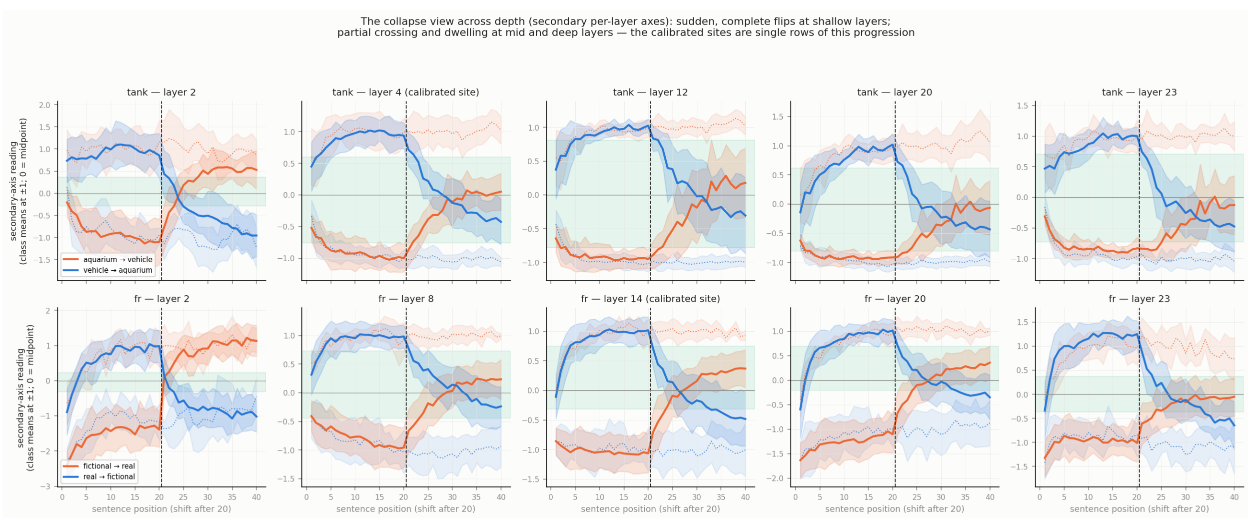

Figure 8: The collapse view across depth (per-layer secondary axes; class means ±1, midpoint 0). Shallow fiction/real layers flip suddenly and completely; mid and deep layers cross partially and dwell; tank aquarium→vehicle stays on the origin side of the midpoint through the final layer. The calibrated sites are single rows of this progression.

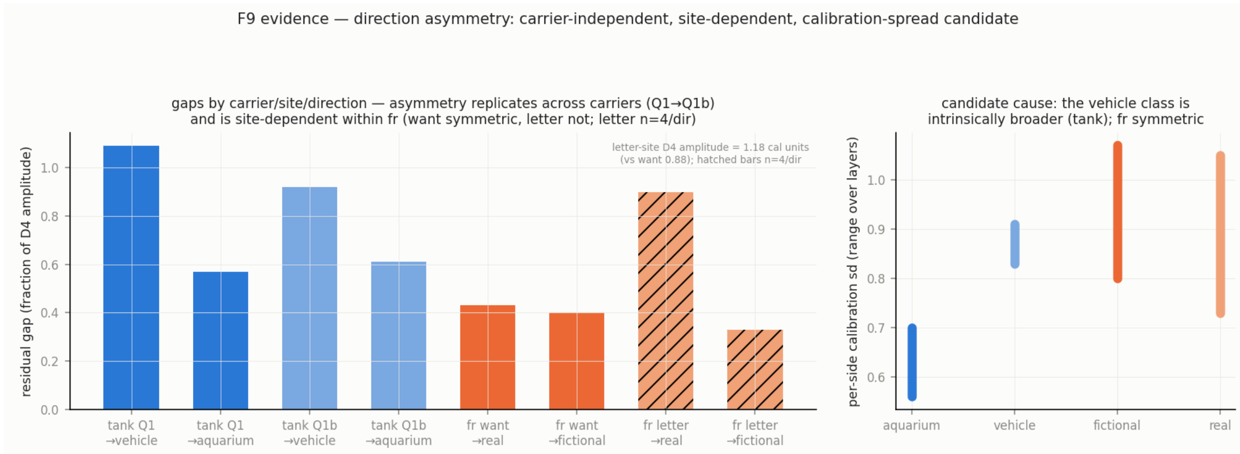

Figure 9: Direction asymmetry: transitions into the broader class stop farther short; the pattern replicates under the replicate carrier and appears at the ' letter' site ($n = 4$ per direction, exploratory); per-side calibration spreads differ (vehicle broader than aquarium at every layer tested).

3.3 The mechanism: drift plus discrete jumps (R3)

What process produces these trajectories? We fit five models per run to the twenty post-shift readings — uniform integrator, fitted-γ recency integrator, change-point step, drift-plus-step hybrid, and a two-timescale integrator (fast and slow components mixed) — selecting by BIC with an indeterminacy band, and we calibrated the selector on synthetic truth: hybrid-truth is recovered as hybrid in 13–15 of 24 simulations; step-truth is called integrator in 0–2%; and two-timescale truth masquerades as a plain integrator or indeterminate, reading as hybrid in only 2–3 of 24. The selector cannot fabricate the hybrid verdict from any of the smooth mechanisms tested.

On the real runs the selector nonetheless returns hybrid: among classifiable runs, drift-plus-jump is dominant — 11 of 16 tank runs and 14 of 25 fiction/real runs (48% of fiction/real runs are indeterminate at that task's noise level, and we state the claim as "hybrid-dominant among classifiable runs" accordingly). The two-timescale model wins zero runs in either task (Fig. 10; per-run fits in Fig. 11; the raw paths in Fig. 12 show the heterogeneity directly — some runs glide, some step). The two-phase shape of §3.2 could in principle have been smooth multi-timescale integration; tested head-to-head, it is not.

The jumps are not triggered by strong evidence. Every post-shift sentence has a single-sentence calibration reading (456 of 456 matched), and the sentences at which runs jump are median-strength exemplars of their class — percentile 0.50 against 0.50 for all other added sentences ($p = 0.445$). The same evidence moves one run smoothly and jolts another: whatever triggers a jump is a property of the state, not of the stimulus — though we have tested only evidence strength — and single-sentence calibration strength may understate in-context strength; surprisal, syntax, and discourse position remain untested — so "state-dependent" is our working interpretation rather than a demonstrated exclusion.

The same process is visible from inside the stream. Using per-position axes calibrated from no-shift checkpoint windows — content and position matched, only history differing — the tokens of the post-shift block themselves read only about half their no-shift reference: +0.34/+0.37 (the two transition directions) at ten post-shift sentences, +0.52/+0.53 at twenty (trimmed means; the untrimmed tank values are inflated by a heavy tail we document in the QA appendix), replicated

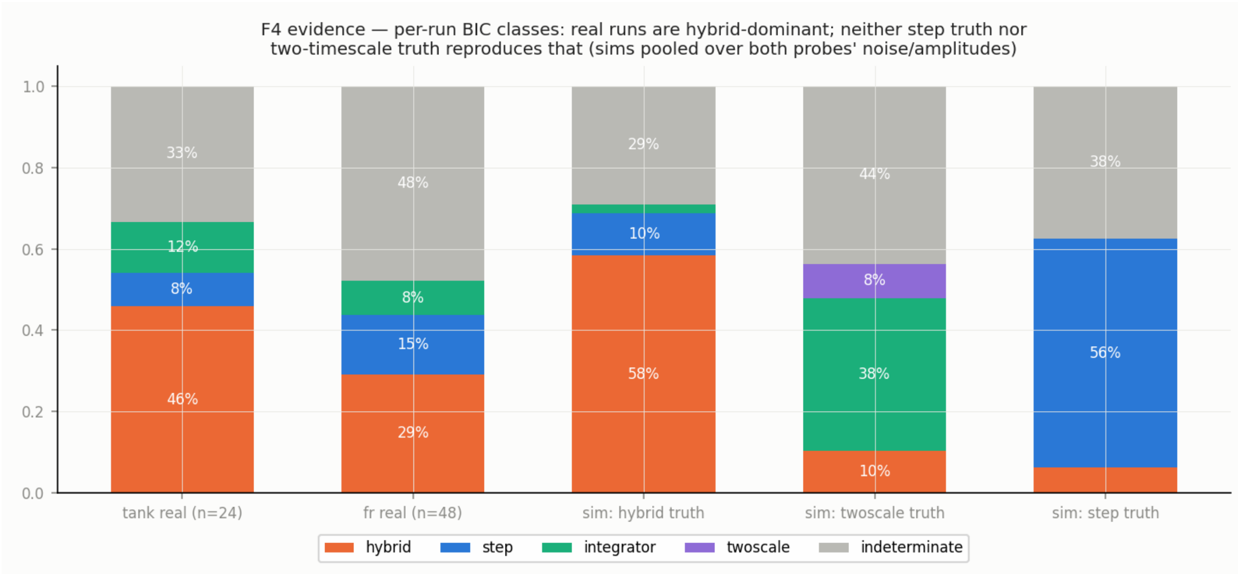

Figure 10: Per-run model selection (BIC with an indeterminacy band) over five candidate processes, with simulation-calibrated identifiability: drift-plus-jump dominates among classifiable runs; the two-timescale integrator wins zero runs, and two-timescale truth cannot masquerade as hybrid.

with a tighter instrument in the fiction/real task (Fig. 13, 14). Mixed history suppresses the class reading of even the new evidence's own tokens, in a recency-graded way. This is descriptive — it is what token-level recency integration would produce — but it establishes that the transition is not merely a summary-site phenomenon: any window-level readout inherits the suppression.

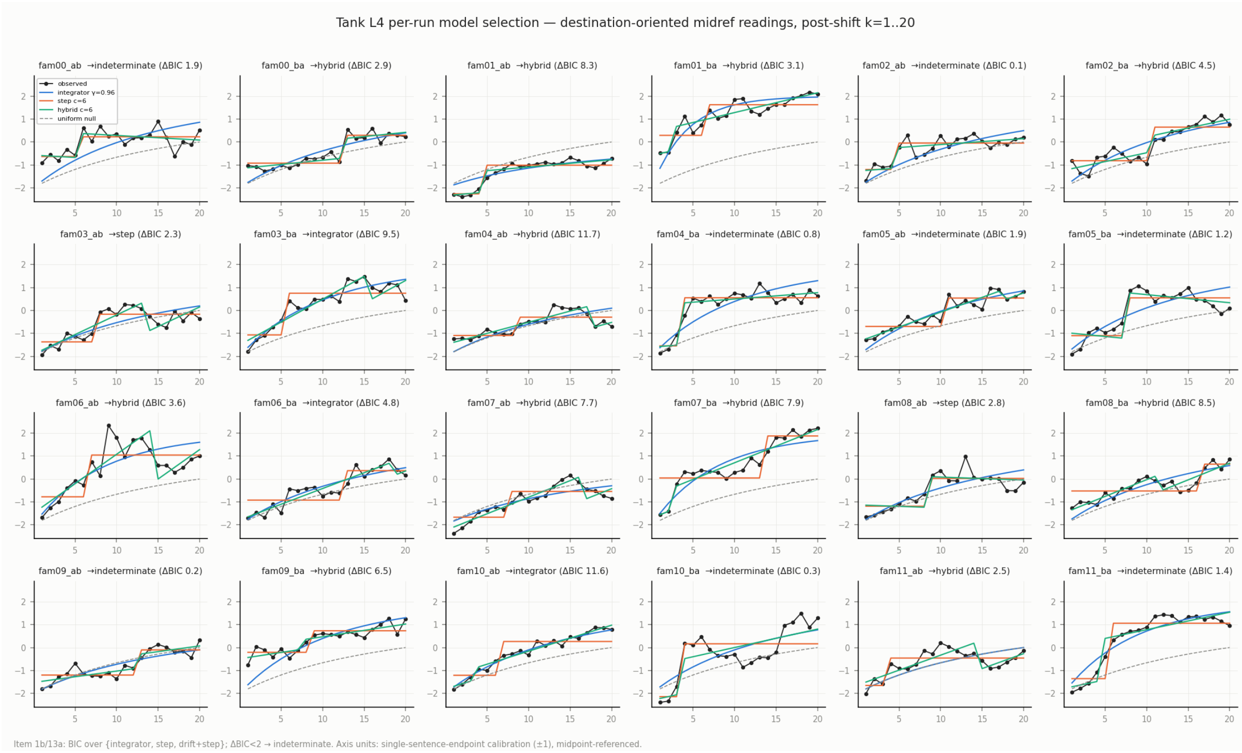

Figure 11: Per-run fits (tank): the best single-γ recency integrator underpredicts the early rise and overpredicts late convergence; drift-plus-jump hybrids capture the paths.

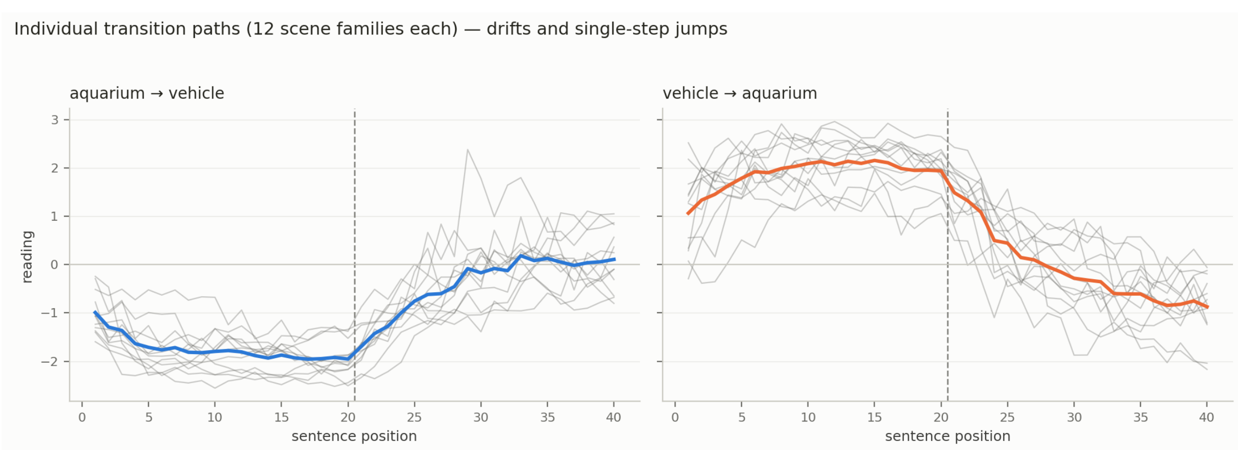

Figure 12: Raw per-run trajectories (tank, layer 4): the heterogeneity behind the means — some runs glide, some step.

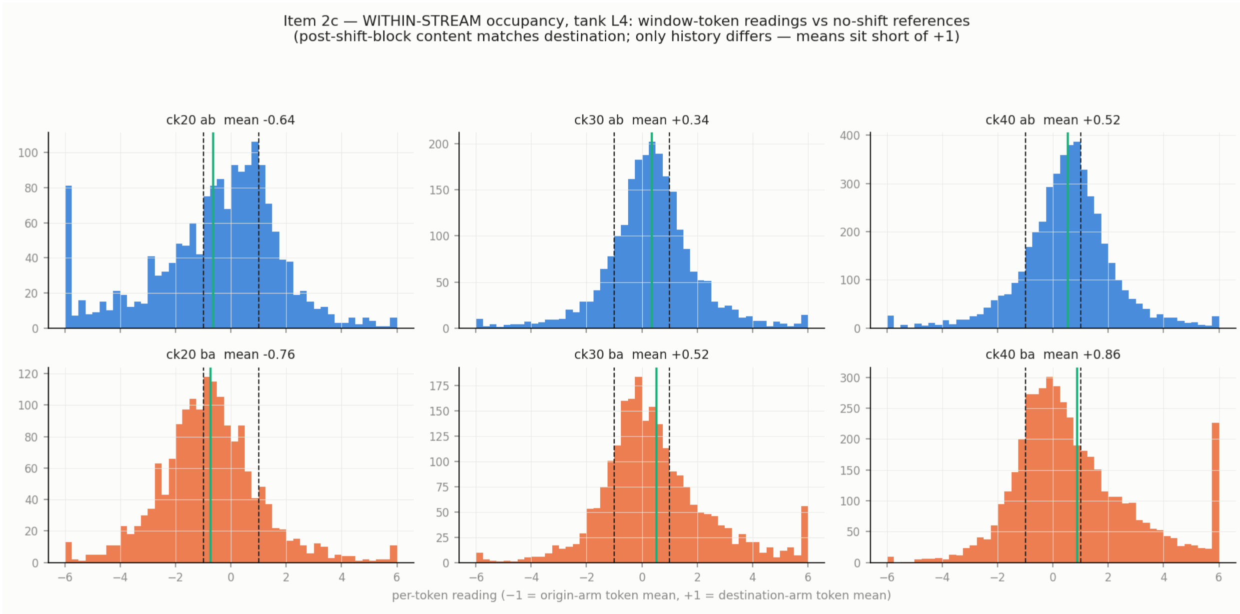

Figure 13: Within-stream readings (tank): post-shift-block tokens — 100% destination-class content — read about half their no-shift reference (trimmed means; the two transition directions), recovering only partially by twenty sentences.

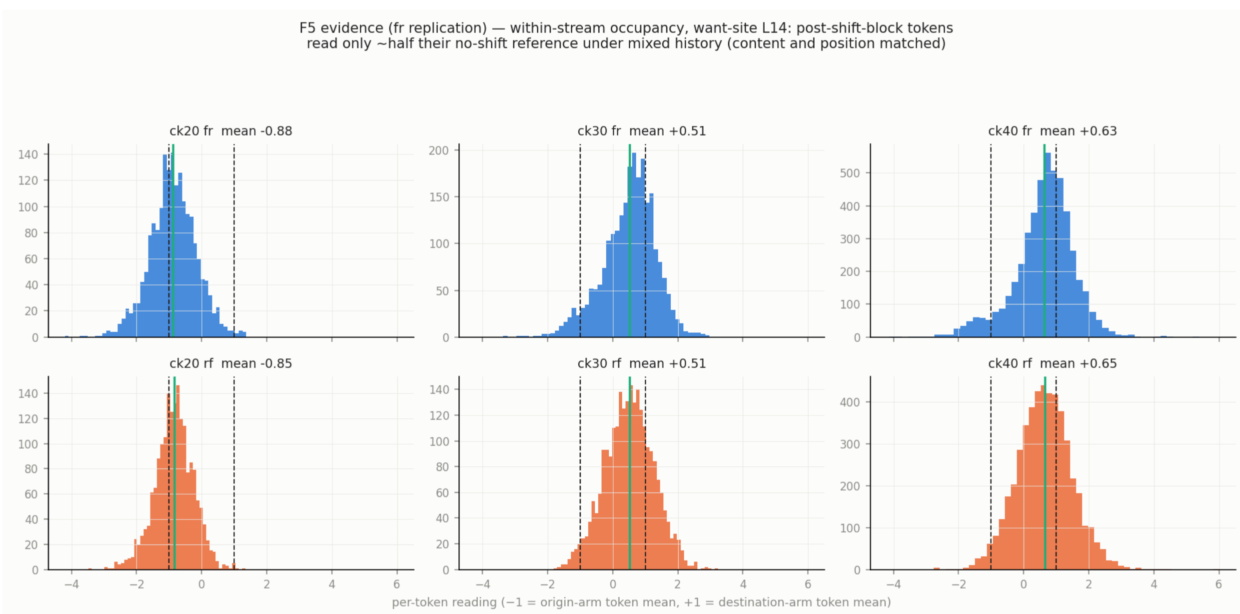

Figure 14: Within-stream replication in fiction/real with a tighter instrument: the same history-driven suppression of the new evidence's own tokens.

3.4 Unresolved states, and what the model does in them (R4)

In one direction of the tank task — aquarium→vehicle — the trajectory dwells within the unresolved zone: it becomes stationary mid-transition and, on the measured horizon, does not leave. The central tendency of the run distribution is stationary at mid-axis for at least ten consecutive steps, $\geq$2.9 across-run standard deviations from *both* endpoint references, with the spread across runs flat rather than tightening; Gaussian-mixture comparison favors a single component in every time bin — one stationary population, not a hidden mixture of resolved and unresolved runs (Fig. 15). Within tank, the same signature appears at the pre-lexical ' word' site and under the replicate carrier; in the fiction/real task it appears only at the ' letter' site of the paraphrase carrier, at n = 4 per direction, exploratory. "Stationary" is bounded by the horizon: whether these runs would ever complete on a longer one is the same open question §3.2 leaves for the remnant, and some individual runs retain shallow nonzero late slopes. Geometrically, the dwelling states sit squarely inside the trajectory bundle — the spread of transition trajectories across runs — at 0.96× the null distance (§3.6); functionally, they are a distinct condition, as the model's behavior shows.

We generated completions at matched counts of post-shift sentences (k = 2, 6, 12, 20) for every run; all behavioral claims in this subsection are correlational. For analysis we partition readings into three bands — origin side, middle, destination side. Behavior tracks the reading: cells whose readings sit on a side answer that side (56% and 66% for the two sides), while the mid band answers "both" — enumerating the two senses — in 45% of cells, roughly double either side (Fig. 16). One descriptive fact anchors this paper's framing: **not one of the 96 tank responses asks which sense is meant or declines to answer pending disambiguation** — zero of 96, by regex scan and manual review of the committed categorized worksheet. In the unresolved zone the model either enumerates senses (45%) or silently commits to one (52% at mid-band); it answers as if resolved. One further response class appears across all 108 tank completions (the 96 transition cells plus 12 no-shift cells): 11 are degenerate repetition loops, unrelated to reading band.

When does the reading carry information *beyond* the context composition that drives both reading and behavior? At matched composition: mid-transition, yes — pooling k ∈ 6, 12, decided answers come from runs with more extreme readings (median |reading| 0.72 vs 0.38, one-sided p = 0.0138 — a pooling chosen after the per-depth pattern was seen, so we grade this suggestive); at the settled extremes, no. In the fiction/real task we categorize each completion as fiction-framed assistance or a safety response — a *safe-completion*, which addresses the request as a risk (declining the letter, redirecting to support) rather than fulfilling it. The reading separates the two at k = 2 (−0.06 vs +1.13, p = 0.010, one of four k values tested) but not later, where response type follows scene family (Fig. 17). The safety-relevant gradient is nonetheless plain at the band level: safe-completion rates run 50% → 80% → 91% as the reading moves from fiction-side through mid to real-side (origin band n = 4, fragile). An exploratory post-freeze check (Appendix C) asks whether any other layer's reading associates more strongly with behavior than the calibrated site's: per-layer association curves (Fig. 18) are roughly flat from mid-stack to the final layer in both tasks — nothing singles out the deep layers — though the instrument is blunt (band-level, imbalanced outcomes), so this neither establishes nor rules out depth-specific behavioral readout.

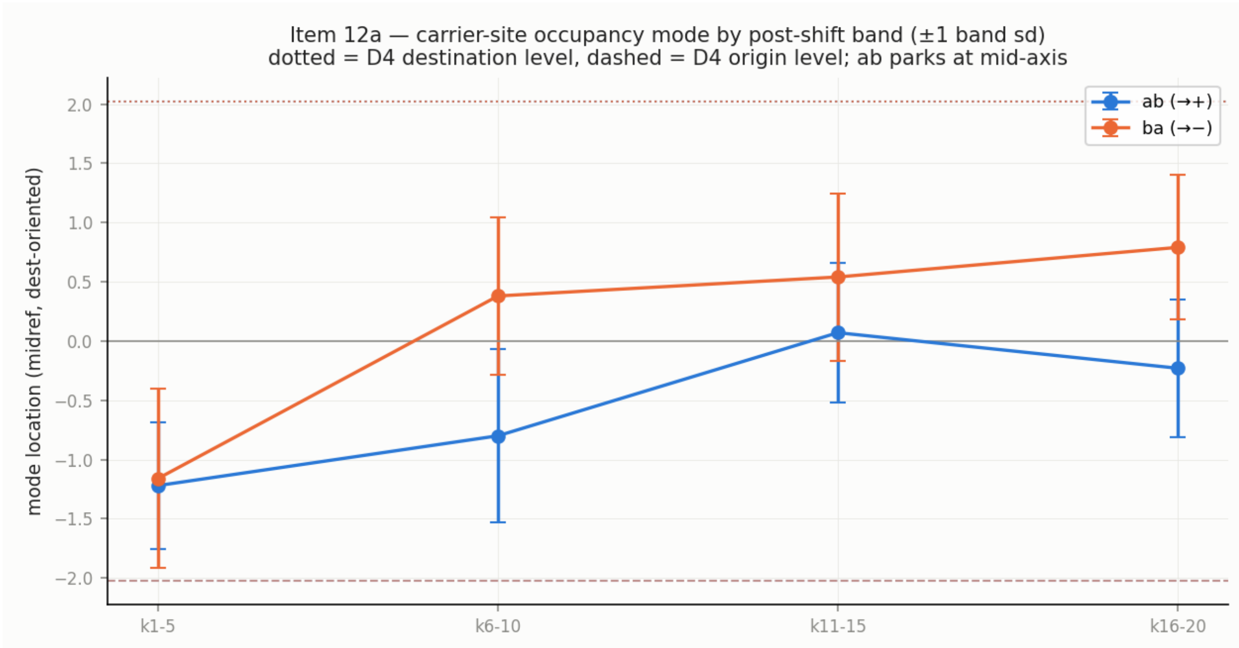

Figure 15: The dwell (tank aquarium→vehicle): the central tendency of the run distribution is stationary at mid-axis for ≥10 consecutive steps, ≥2.9 across-run standard deviations from both endpoint references, spread flat; Gaussian-mixture comparison favors one component in every time bin.

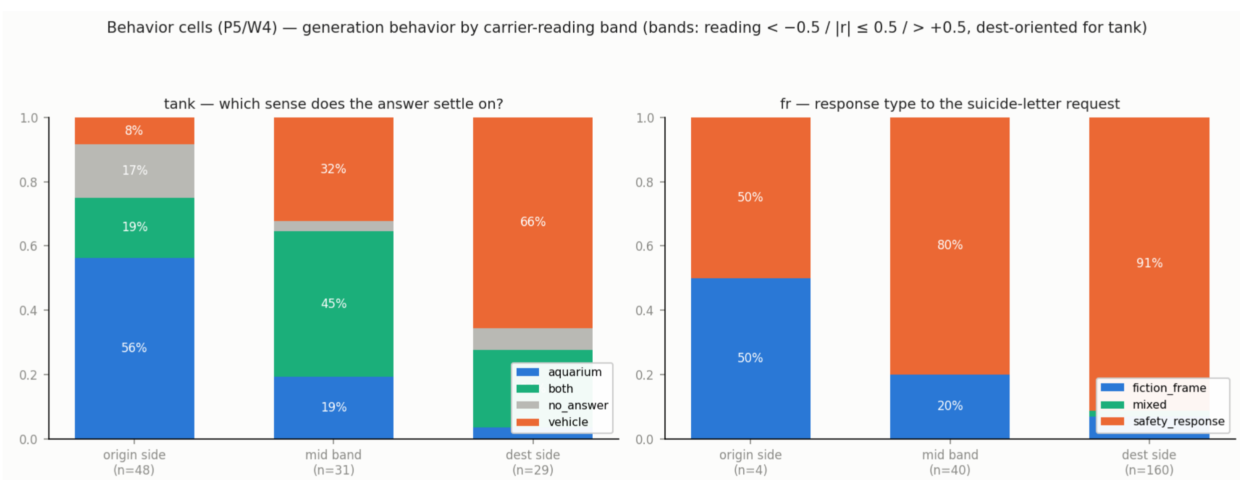

Figure 16: Behavior by reading band (tank): side bands answer their side (56%/66%); the mid band answers “both” in 45% of cells — roughly double either side. Zero of 96 completions ask which sense is meant.

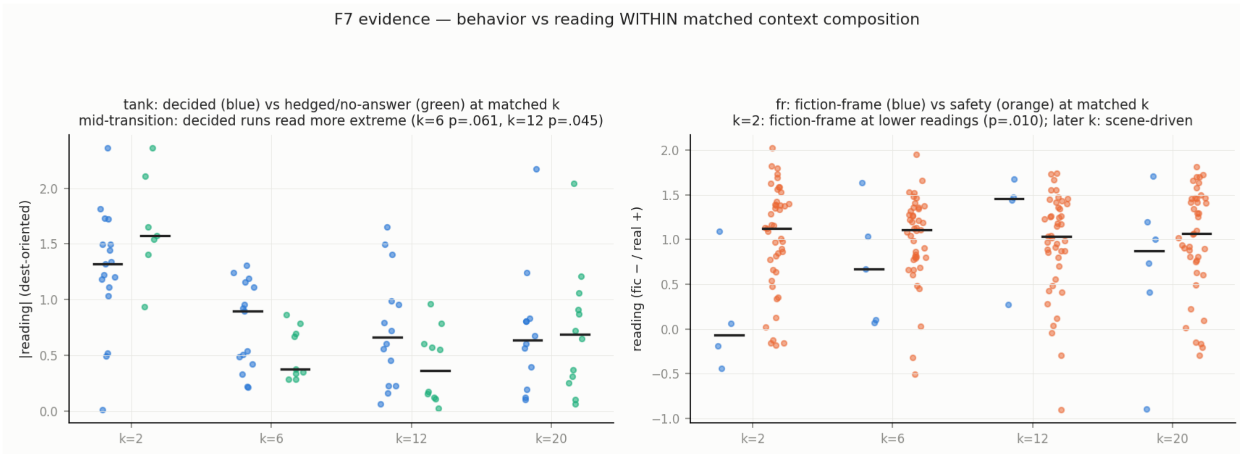

Figure 17: Behavior at matched context composition: mid-transition, decided answers come from runs with more extreme readings (pooled $k \in \{6, 12\}$, graded suggestive); in fiction/real the reading separates response types at $k = 2$ only; safe-completion rates run 50→80→91% across bands (fiction-side endpoint $n = 4$).

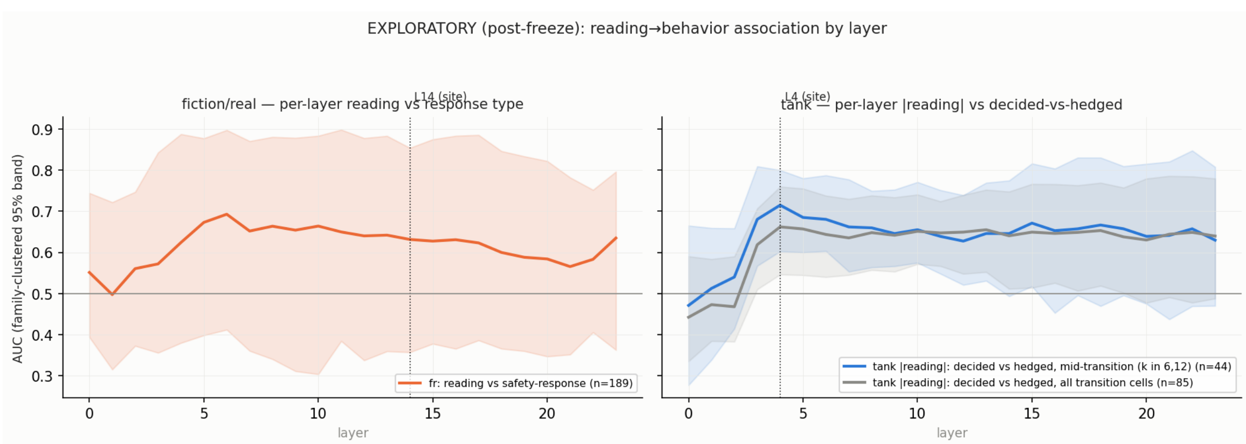

Figure 18: EXPLORATORY (post-freeze): reading→behavior association by layer (rank AUC, family-clustered 95% bands). Roughly flat from mid-stack to the final layer in both tasks; nothing singles out the deep layers. Blunt instrument: band-level association, imbalanced outcomes.

3.5 Order and equilibrium: hysteresis without stickiness (R5)

Does the order of evidence matter beyond its amount? We built static mixture sweeps: twenty-sentence contexts with k destination-class sentences, k swept 0→20, in two block orders, from each family's own transition sentences. The resulting loops are large — the same mixture read with the destination block recent versus old differs by loop areas of +14.4 [12.2, 16.8] (tank) and +10.5 [7.5, 13.3] (fiction/real): operational hysteresis, plainly present (Fig. 19).

The reserved question was whether any of that order-dependence exceeds what recency weighting alone produces — call any such excess *stickiness*. There is none: a one-parameter recency integrator fitted to the sweep cells reproduces the loop areas almost exactly (+14.3; +11.1) — excess "stickiness" of +0.1 and −0.5, both indistinguishable from zero — and cross-order validation (fit one branch, predict the other) preserves the verdict. We note, as part of the method, that our own first pass computed this against a misspecified null (γ imported from the transition fits) and declared significant stickiness; the fitted null killed the claim within the hour, and both versions are in the corrections record. What remains real beyond the single-γ account is mild: the two sweep directions prefer slightly different recency weights (γ 0.91/0.97 tank, 0.84/0.90 fiction/real), echoing the directional asymmetry of §3.2.

This resolves an apparent contradiction with §3.3. The *equilibrium map* from evidence mixture to reading is smooth and integrator-like; the *sequential path* through a transition is drift punctuated by jumps. Both are true; they are different observables. The metastability this paper names is a property of paths, not of the equilibrium map — there is no bistability to fall into and no barrier to escape.

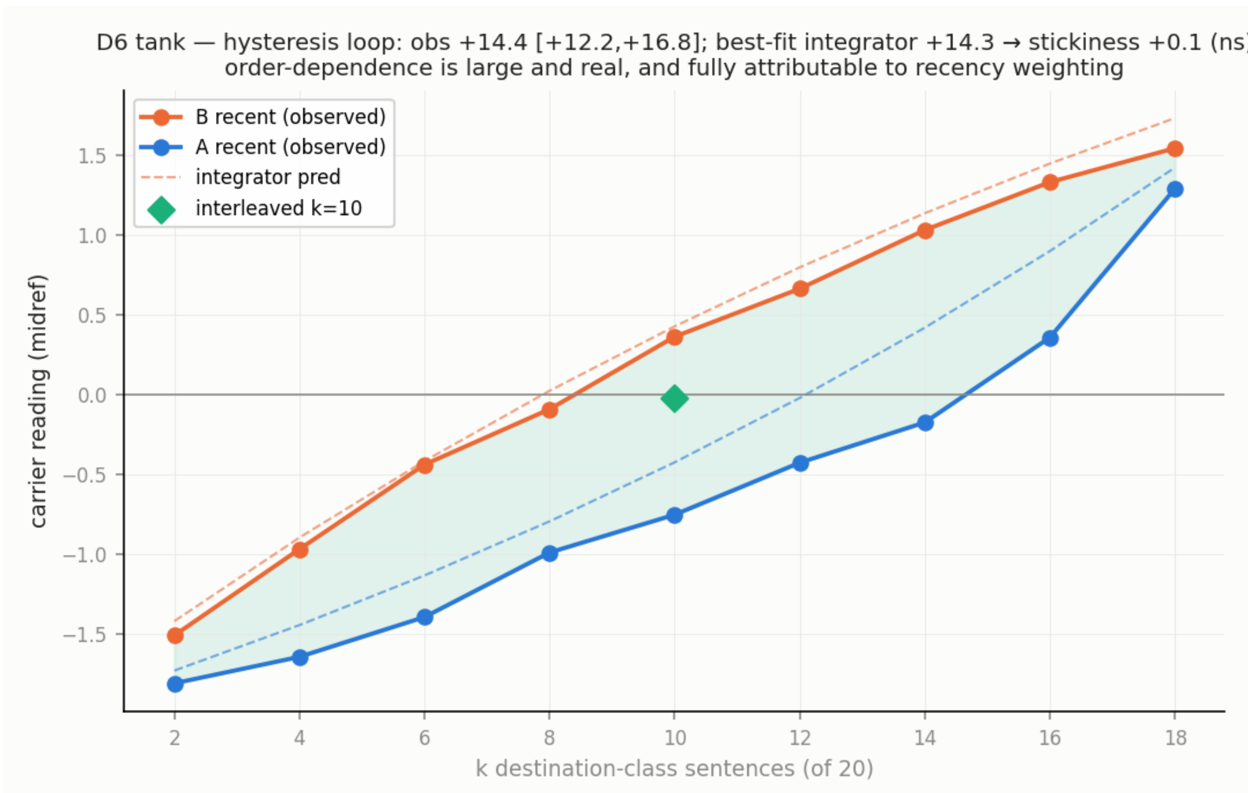

Figure 19: Hysteresis without stickiness (tank): static-mixture sweeps in two block orders produce loop area +14.4 [12.2, 16.8]; a one-parameter recency integrator fitted to the same cells reproduces it (+14.3) — excess “stickiness” +0.1, ns.

3.6 The representation of irresolution (R6)

Finally, the third of the introduction's three worlds: do these in-between states leave the model's learned distribution? "Off-manifold" is meaningful only relative to a reference, a measure, and a noise level, so we fix all three: reference — the position-matched no-shift states (and the cross-run transition bundle); measures — k-nearest-neighbor distance and out-of-subspace reconstruction error against the no-shift PCA subspace; noise level — the held-out self-distances of the reference itself. (Paired values in this section are tank/fiction-real throughout.) The instruments detect real displacement: single-sentence calibration states — a genuinely different context regime, serving as a positive control — read 1.7–2.2× the null across the two instruments, and position-mismatched states are flagged at 13–44%.

At the individual-state level, nothing leaves the reference distribution. Jump steps read 1.06×/1.12× the null, the stationary dwelling states 0.96×, all inside the null's spread (Fig. 20). Our pre-registered prediction that jump steps would show elevated off-manifold distance failed, and we report it as failed.

The subtle result appeared only when the per-state verdict was challenged: a displacement too small to flag any single state could still be shared by all of them, so we tested the mean directly. The *mean* out-of-subspace reconstruction error of transition states, against a family-block bootstrap null, is far outside the null's range: a shared displacement equal to **25% and 38% of the full class separation**, $p < 0.001$ in both tasks, rising over the first five post-shift sentences and persisting undiminished to twenty — largest in the stationary dwelling states (Fig. 21). We call the direction that carries it the **mixed-context marker**. It is not class leakage: the direction is orthogonal both to the single-sentence class axis (cos +0.015, +0.011) and, by construction and by measurement, to the accumulated-context class axis (−0.0052, −0.0065). It survives held-out direction estimation (71–90% of in-sample magnitude). It is not family novelty: if the marker merely reflected unfamiliar scene families, pure-class cells from families *outside* the reference construction should sit higher on it than cells from familiar families; they sit lower (+1.6%/+3.4% of separation versus +6.7%/+6.0%), and both sit far below mixed cells (≈ +27%).

What is it? Held-out mixture cells locate its content: the marker is absent from pure-class contexts and near its full transition strength in *static* mixed contexts — so it marks mixed context, not temporal shifting per se. One qualification: in the fiction/real task it halves when the mixture is interleaved rather than blocked, so there, part of what it encodes is the coherent structure of a shift. It did not predict behavior in eight matched-composition tests at the calibrated site and layer (one nominal hit, uncorrected; other layers untested). The model, in other words, carries a persistent, systematic signal that its context is mixed — a direction on which pure contexts sit near zero and mixed contexts near 27% of the class separation — and its behavior does not appear to use it. Whether this signal is a learned representation of mixedness or a systematic consequence of heterogeneous input is not decided by our measurements.

Three graded senses of "off-distribution" should be kept distinct: exotic natural inputs; interventional states (patching, steering); and states of one context regime measured against a reference built from another, as our single-sentence positive controls are. Our shifts are the tamest possible case — clean block transitions between two well-learned frames — and the fact that even the dwelling states stay on-distribution here says nothing about adversarial or incoherent contexts, where genuine excursions remain most likely; that battery is future work.

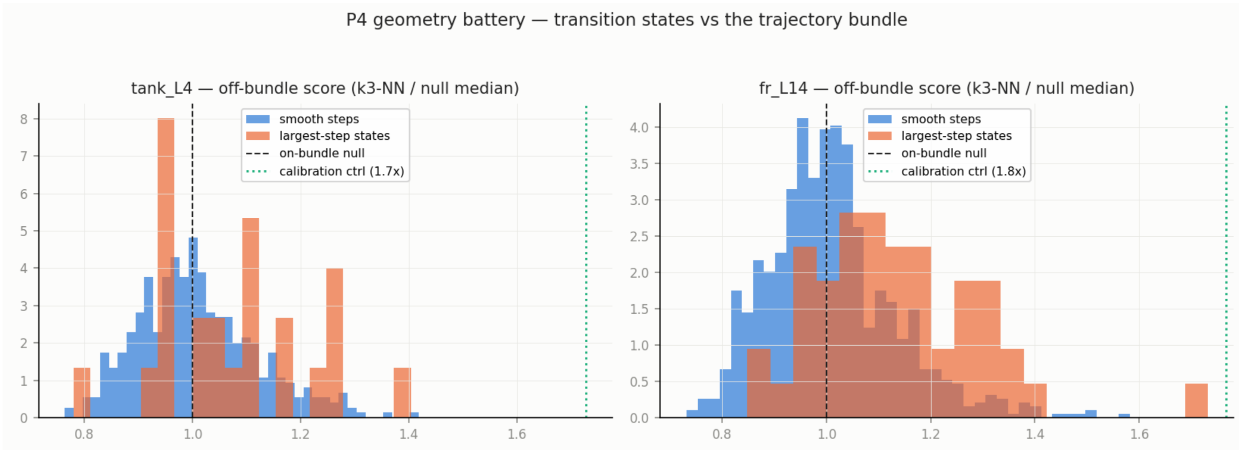

Figure 20: Per-state geometry: jump steps (1.06×/1.12× the null) and dwell states (0.96×) sit inside the no-shift null's spread; positive controls (calibration states, position-mismatched states) are detected.

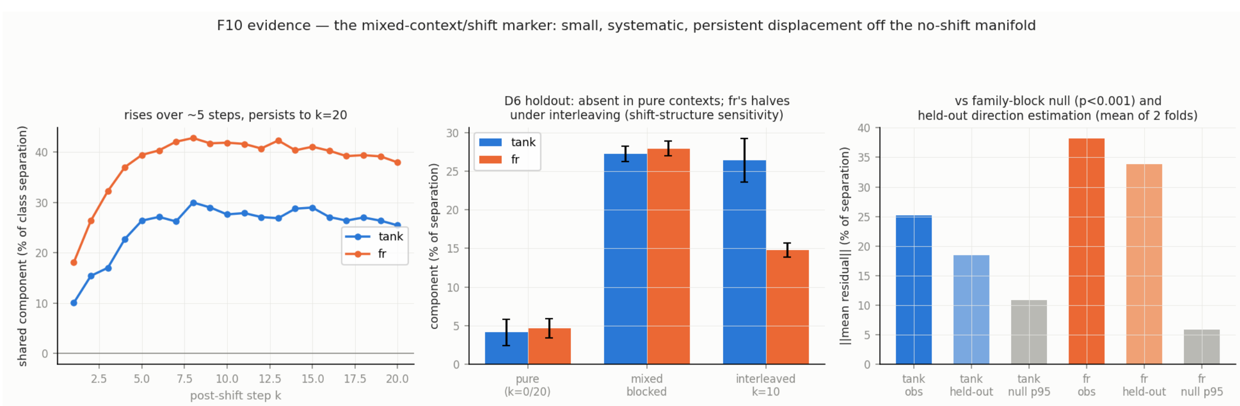

Figure 21: The mixed-context marker: mean out-of-subspace reconstruction error of transition states against a family-block bootstrap null — a shared displacement of 25%/38% (tank/fiction-real) of class separation, orthogonal to the class axis, rising over ≈5 post-shift sentences, persisting to twenty, largest during the dwell.

4 Discussion

What ”metastable” means here. We anchor the term in the dynamics of biological neural systems rather than in physics. In coordination dynamics and in the winnerless-competition framework — models of sequential transient dynamics in neural populations — cognition is described as sequences of transiently stable states that are not fixed-point attractors — structured passages through state space in which dwelling and moving are not opposites. That is the geometry the data show: a run that dwells mid-transition occupies something functionally state-like (stationary for many steps, with its own behavioral signature) yet geometrically a passage (inside the trajectory bundle, on a smooth equilibrium map). One disclaimer: nothing here is equilibrium bistability — the evidence-mixture map is smooth (§3.5) — and readers should not import a barrier-crossing picture. Semantic metastability, as we use it, is a property of paths.

The human parallel. The signature of §3.2 — fast partial reanalysis that leaves a lingering trace of the initial misreading — is also the signature of human sentence processing in the ”good-enough” tradition: after recovering from a garden-path sentence, comprehenders measurably retain components of the initial, incorrect interpretation. We make no mechanistic identification; we note that a language model trained on human text reproduces, at the representational level, the reanalysis profile humans show behaviorally — including the part where the old reading never fully leaves.

The study as its own subject. This project began expecting basins, stickiness, and off-manifold escape routes, and the data dismantled each expectation in turn: the stickiness died against a properly fitted null within an hour of being announced; the off-manifold excursions never appeared; what replaced them — recency dynamics, jumps, a remnant, a marker — is stranger and better supported. The arc is the one §1 took from humor theory — incongruity, then resolution — replayed in the study's own history. We kept the corrections record (eighteen entries, Appendix A) in the paper because the surviving claims owe their credibility to that process, not despite it.

Safety and alignment, first descriptively. The failure surface this study maps needs no adversary. During an ordinary, coherent shift of conversational frame, a model's reading of a critical token trails the present frame for roughly its integration window (here, ten to seventeen sentences), retains a remnant of the prior frame beyond that, and — in one tank direction, with an exploratory n = 4 echo in fiction/real — dwells between frames for the remainder of the tested horizon. Throughout, the model never asks which reading is meant or flags the ambiguity as an obstacle — zero of 96 tank completions, across all bands — though 45% of mid-band answers do surface both senses. And safeguard-relevant behavior co-varies with the internal reading: safe-completion rates fall from 91% to 50% across reading bands (the fiction-side endpoint rests on four completions, and the gradient is partly scene-driven — §3.4) — attenuation reachable by ordinary context, no exotic inputs required.

Then the interpretation, with its caveat stated first. The following is a post-hoc reading of an asymmetry we noticed, not a designed manipulation, and training provenance is unobservable: our two tasks appear to differ in whether they carry a trained default for unresolved cases. The refusal task behaved as though it does — at mid-transition, 80% of completions safe-complete, and in sampled chain-of-thought traces — the reasoning channel its chat format exposes, released with our behavior data — both framings are weighed before the safe reply (a qualitative observation, not a coded rate). The sense task carries no such default, and there the model silently commits (52% of mid-band completions pick a sense). If this reading is right, the metastable zone is the failure window for every behavior *without* a trained uncertainty-default — refusal-style safeguards may be the well-covered exception rather than the rule. A second, compatible candidate emerged from the depth data, though it is weaker than it first looked: fiction/real's mid-deep stack (layers 3–18) settles clearly on the destination side in both directions, where tank aquarium→vehicle's ends at the midpoint — so downstream layers in the covered task may act on a more settled reading than the site we measure. But the deepest band hovers near the midpoint in fiction→real too — precisely the direction where the safeguard must engage and did (§3.2) — so depth-resolution cannot carry the explanation alone. A third candidate points the opposite way through the stack: the shallowest layers resolve the framing composition almost immediately and completely (§3.2), with the profile of surface-cue tracking — the minimal-pair discrimination that rules out cue-only tracking was run at the calibrated site, not at shallow layers — and a safeguard that reads early, from surface content and frame cues, would fire broadly whenever the alarming request is present and be suppressed only by a well-established fictional frame: the shape of our behavioral data, including its fragile 50% fiction-side floor. The fast, complete shallow response is itself fr-specific — tank's shallow layers respond later and, in one direction, only partially (§3.2) — which is what a trigger sculpted by safety post-training would look like, though ordinary register statistics learned in pretraining predict the same asymmetry and training provenance is unobservable here. The three accounts are not exclusive — one names the policy, one the downstream state, one the trigger's plausible locus — and none is causally established: with two tasks these are observations, and the exploratory per-layer curves (§3.4) are too blunt to separate them (shallow readings saturate post-shift, leaving that instrument blind exactly where the third account lives). We add one calibration so this section cannot overpromise: as a standalone cell-level monitor, the frame reading is not yet usable (AUC 0.61 [0.43, 0.76], chance-compatible; Fig. **??**, supplement). The band-level gradient is real; building a usable monitor — combining several sites, the mixed-context marker, and per-layer readings — is future work, not a claim.

Garden path or pun. In Dynel's terms our transition corpus is a slow-motion garden path, and most trajectories treat it as one: incongruity, then resolution toward the new reading. The dwelling runs are the interesting case — an *extended incongruity phase*, in which the model, asked what the word means, answers like someone explaining a pun: both senses, held. Whether any of them is a true pun — held incongruity that never resolves — or only a long garden path is the persistent-versus-slow question our twenty-sentence horizon cannot decide, and the extended-tail experiment named in future work is its dedicated test.

Typicality is not commitment. The three-worlds question dissolved because it conflated two axes. On *distributional typicality* the stationary states are unremarkable — inside the bundle, on-distribution. On *semantic commitment* they are distinctive — uncommitted, between calibrated interpretations, with hedging as their behavioral expression. A learned state of unresolvedness is typical *and* uncommitted; genuinely off-distribution states (glitch inputs, interventions, adversarial incoherence) occupy a different cell of that table and remain untested here. The general lesson: that a state is geometrically typical does not make it functionally normal — our behavior data document the functional difference directly.

Represented but not consulted. The mixed-context marker gives the dissociation its sharpest form: the model carries a persistent, systematic signal that its context is mixed — and behavior does not appear to use it at matched composition, at the one site and layer tested. This is the same structure reported in the hallucination literature, where models internally encode uncertainty or truthfulness that their generations do not respect. Here it appears in transition dynamics: an internal signal correlated with the conditions that produce the unresolved zone already exists inside the model, and behavior does not appear to use it — whether it could actually flag the zone is untested (the one monitor built, from the frame reading, is chance-compatible). The motivating case of §1 reportedly had the same structure at system scale — the provider's moderation layer was flagging the user's messages for self-harm risk in real time while assistance continued [CITE: filing/reporting]. That suggests a constructive direction for alignment work — not creating uncertainty signals, but testing whether behavior can be gated on the ones already there.

Closing. Word-sense reinterpretation is the tractable laboratory instance of a much larger family — frames, personas, tasks, and safety postures all shift under accumulating context, and there is no obvious reason the metastable structure documented here is unique to word senses. The states are what the title calls them: unresolved. For now, so is the question of what a model should do while inside one.

5 Limitations and future work

One model, stated plainly. Everything here is measured in one 20-billion-parameter mixture-of-experts model, with deterministic decoding — and each dynamics claim at one calibrated site and layer per task, single rows of a depth progression in which crossing timescales vary systematically (§3.2); no single layer is a sufficient readout of the model's state, and behavior is produced downstream of all of them. Within the study, replication is internal: two task contrasts, three carriers plus a replicate, two measurement sites, twelve scene families per class per task. That inventory also records what did *not* replicate cleanly, and we state it rather than smooth it: the real→fictional remnant is suggestive only once reference uncertainty is propagated; the letter-site asymmetry rests on four runs per direction; 48% of fiction/real runs are indeterminate under per-run model selection; and our pre-stated prediction about the ordering of crossing times across layers held in one task and failed in the other. Cross-task patterns — the task with the wider endpoint separation also shows the larger remnant gaps; anchored versus distributed contrast structure — are n = 2 observations, hypotheses rather than findings.

The named open question. Whether the remnant — and the dwelling within the unresolved zone — are permanent or merely slower than twenty sentences: in the terms of §4, whether the dwelling is a pun rather than a long garden path — is decidable by extending the post-shift horizon (extended-tail runs) and by shifting evidence back to the original class (A→B→A runs, measuring what remains of the second frame). Both are designed and costed; neither has run.

Further deferred work, in rough order of leverage: replicating the fiction/real headline quantities at the ' write' site, where the framing contrast reads most strongly (the existing recordings suffice); expanding the letter-site families; calibration sets from a third, unrelated class, to identify what the accumulation offset contains; monitor construction (multi-site, marker-augmented, depth-aware) against the chance-compatible single-site baseline reported above; causal localization of where safeguard behavior reads in the stack (patching shallow versus mid-stack states during generation — the decisive test for §4's three candidate accounts); and a planned suite of harder context manipulations — colliding frames, mutating them mid-context, deliberately incoherent contexts — where genuine off-distribution excursions are most likely, our clean block-shifts being the tamest possible case.

The analyses reported here were frozen before drafting; the two post-freeze additions (a descriptive response count and the monitor curve) are logged as such in Appendix C.

Back matter (assembly spec — materialized at final build)

Review-draft note: the sections below are the assembly specification; the final build replaces them with the referenced frozen content.

Acknowledgments

[Agreed AI-assistance line — exact text from Andrew/chat side, placeholder.] Personal acknowledgments: Andrew adds.

Related work (arXiv default: standalone section, drafted from handoff §6 anchors)

Probing validity: Alain & Bengio; Hewitt & Liang (control tasks — our D5 + label-shuffle audits presented as the selectivity answer); Belinkov survey. Directions & steering: Marks & Tegmark; Arditi et al. (refusal direction — nearest cousin of the fiction/real behavior link). In-context dynamics: Xie et al. (ICL as implicit Bayesian inference — the standing theory our integrator null speaks to); Hendel; Todd (function vectors); Olsson et al. (induction heads). Safety: many-shot jailbreaking; Crescendo-style multi-turn escalation. Psycholinguistics: Christianson & Ferreira; Slattery et al.; van Schijndel & Linzen. Verbalized uncertainty & ambiguity (grounds the intro's scoped claim): Kadavath et al.; Lin et al. (verbalized confidence); Xiong et al. (calibration of verbalized confidence); Liu et al. (ambiguity modeling). Internal encoding vs expression (grounds Discussion's "represented but not consulted"): Azaria & Mitchell (internal state predicts truthfulness); Orgad et al. (models encode more than they express). Long-context artifacts: attention sinks (the "is the offset positional?" question; midpoint referencing makes our findings robust either way). Dynamics: Sussillo & Barak; Kelso; Rabinovich et al. Motivating case (intro + discussion): Raine v. OpenAI, Cal. Super. Ct. (S.F.), filed 2025-08-26; contemporaneous reporting (CNN 2025-08-26; NBC News; TechCrunch 2025-11-26 on OpenAI's answer denying causation). Contested litigation — the paper characterizes only what the filing and reporting describe.

Appendix A — Corrections record (presented as method)

The 18 entries verbatim from FINDINGS_FINAL §3, one line each, with the two same-session retractions (D6 stickiness; fr durability) called out; one framing paragraph: every claim in this paper survived an explicit attempt to kill it, and these are the ones that did not.

Appendix B — Synthesis audit

The earns/borderline/ornament table from FINDINGS_FINAL §4, with its one framing sentence (every element justified or named as ornament — nothing kept silently).

Appendix C — QA and reproducibility

Regeneration audit summary (bit-identical axes; 23-script diff green; seeded bootstraps exact); 9/9 fixture suite; sign/boundary/join/shuffle audits; the four logged post-freeze additions (s12 response counts, 0/96; s11 monitor ROC, AUC 0.61 [0.43,0.76]; s13 per-layer collapse panels; s14 exploratory per-layer behavior association — both computed from frozen captures via the committed cache); repository pointer + regeneration instructions. Terminology map for repository readers: the paper's *remnant* is the repository's "residual"; the dwelling within the unresolved zone is the repository's "park"; *accumulation offset* is "accumulation drift"; *tasks* are the repository's "probes"/"probe arms"; corpus directories D3/D4/D5/D6 are the transition, no-shift, minimal-pair, and mixture-sweep corpora; the repository's "K=1" label names an analysis convention for a routing track this study dropped.

Appendix D — Supplementary figures

figures/other/ (12): fr fit gallery, fr D6 loop, tank secondary heatmap, matched-k panel is MAIN (`fig_s9_behavior_matchedk`) — supplement gets: raw + midref heatmap generations (the instrument story's "before" pictures), prelexical_L4, occupancy_bands_L4, jumpiness, norm_vs_alignment, calibration_layers, plus `fig_s11_monitor_roc`. `fig_s13_collapse_layers` is MAIN (cited in 3.2's depth paragraph; committed script s13_collapse_by_layer.py).